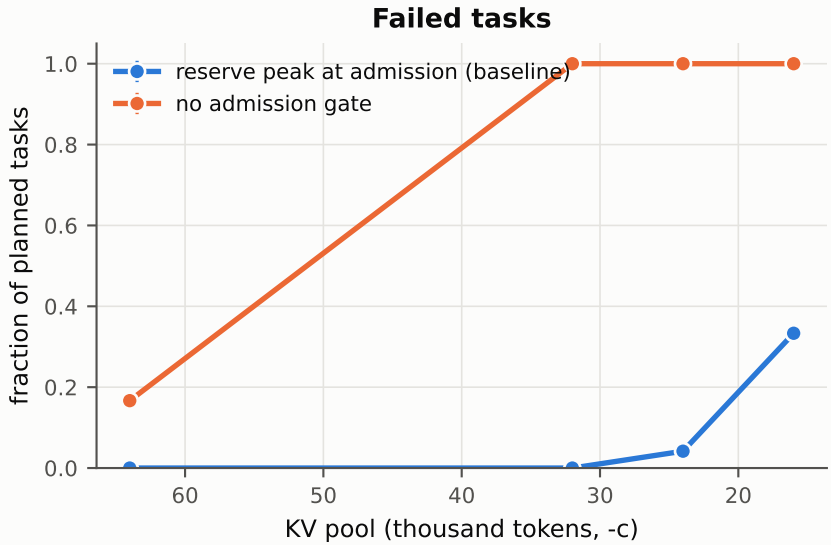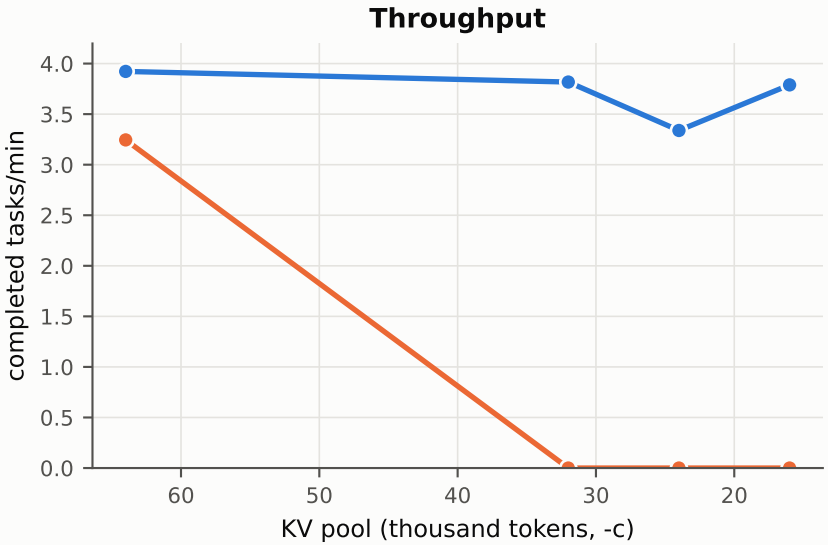E6: shrinking the memory budget (trace_agentic, K=4)